Tensor products of vector spaces

Can combine two (arbitrary) vector spaces V and W to form the *tensor product* $V \otimes W$. The elements of $V \otimes W$ are linear combinations of "tensor products" $|v\rangle \otimes |w\rangle$ consisting of elements $|v\rangle \in V$ and $|w\rangle \in W$.

Example: let $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ be the single qubit spaces with basis $\{|0\rangle, |1\rangle\}$, then

$$\frac{1}{2} \underbrace{|0\rangle \otimes |0\rangle}_{=|00\rangle} + \frac{5i}{7} \underbrace{|1\rangle \otimes |0\rangle}_{=|10\rangle}$$

is an element of $V \otimes W$.

Let $\{|i\rangle_V : i = 1, \ldots, m\}$ be a basis of V, and $\{|j\rangle_W : j = 1, \ldots, n\}$ be a basis of W, then

$$\left\{ |i\rangle_V \otimes |j\rangle_W : i = 1, \ldots, m, j = 1, \ldots, n \right\}$$

is a basis of $V \otimes W$. In particular:

$$\dim(V \otimes W) = \dim(V) \cdot \dim(W).$$

Note: $|i\rangle_V \otimes |j\rangle_W$ is also written as $|i\rangle |j\rangle$ or $|ij\rangle$.

Example: the basis construction for $V = \mathbb{C}^2$, $W = \mathbb{C}^2$ leads to the standard two-qubit basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

Elementary properties of tensor products:

- for all $|v\rangle \in V$, $|w\rangle \in W$ and $\alpha \in \mathbb{C}$:

$$\alpha(|v\rangle \otimes |w\rangle) = (\alpha |v\rangle) \otimes |w\rangle = |v\rangle \otimes (\alpha |w\rangle)$$

- for all $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$:

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle$$

- for all $|v\rangle \in V$ and $|w_1\rangle, |w_2\rangle \in W$:

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle$$

Vector notation (using standard basis), e.g.,

$$|v\rangle = v_1 |0\rangle + v_2 |1\rangle = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \quad |w\rangle = w_1 |0\rangle + w_2 |1\rangle = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix},$$

then:

$$\begin{aligned} |v\rangle \otimes |w\rangle &= (v_1 |0\rangle + v_2 |1\rangle) \otimes (w_1 |0\rangle + w_2 |1\rangle) \\ &= v_1 w_1 |00\rangle + v_1 w_2 |01\rangle + v_2 w_1 |10\rangle + v_2 w_2 |11\rangle \end{aligned}$$

Thus:

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \otimes \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} v_1 w_1 \\ v_1 w_2 \\ v_2 w_1 \\ v_2 w_2 \end{pmatrix}$$

Example:

$$\begin{pmatrix}2\\3\end{pmatrix} \otimes \begin{pmatrix}5\\7\end{pmatrix} = \begin{pmatrix}10\\14\\15\\21\end{pmatrix}$$

Note: not every element of $V \otimes W$ can be written in the form $|v\rangle \otimes |w\rangle$ with $|v\rangle \in V$ and $|w\rangle \in W$, for example the so-called Bell state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Assuming that V and W are equipped with an inner (scalar) product $\langle\cdot|\cdot\rangle$, one can define an inner product on $V \otimes W$ by:

$$\left\langle \sum_j \alpha_j |v_j\rangle |w_j\rangle \,\middle|\, \sum_k \beta_k |\tilde{v}_k\rangle |\tilde{w}_k\rangle \right\rangle \coloneqq \sum_j \sum_k \alpha_j^* \beta_k \langle v_j|\tilde{v}_k\rangle \langle w_j|\tilde{w}_k\rangle.$$